Tongyi Qianwen large model text interaction

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1. Concept introduction

1.1 What is the "Tongyi Qianwen large model"?

"Tongyi Qianwen" is the ultra-large-scale language model developed in-house by the Alibaba DAMO Academy. It is capable of multi-turn dialogue, copywriting, logical reasoning and multimodal understanding. Integrating Tongyi Qianwen into the `largemodel` project gives the robot a powerful "cloud brain" from a leading domestic technology company.

Compared with local models such as Ollama, the **core advantages** of using an online API like Tongyi Qianwen are:

- **Strong all-round capability**: the Tongyi Qianwen model family (such as `qwen-turbo`, `qwen-plus`, `qwen-max`) performs well across all kinds of evaluations and is capable across the board.
- **Close integration with the cloud ecosystem**: as part of Alibaba Cloud, its API is stable and reliable, and it integrates conveniently with the other Alibaba Cloud services such as OSS and Function Compute.
- **API friendly**: the Tongyi Qianwen API uses the standard HTTP RESTful style, which makes the integration simple and intuitive and easier to implement and debug than approaches such as WebSocket.

In the `largemodel` project, setting the `llm_platform` parameter to `qianwen` switches the language model service to Tongyi Qianwen.

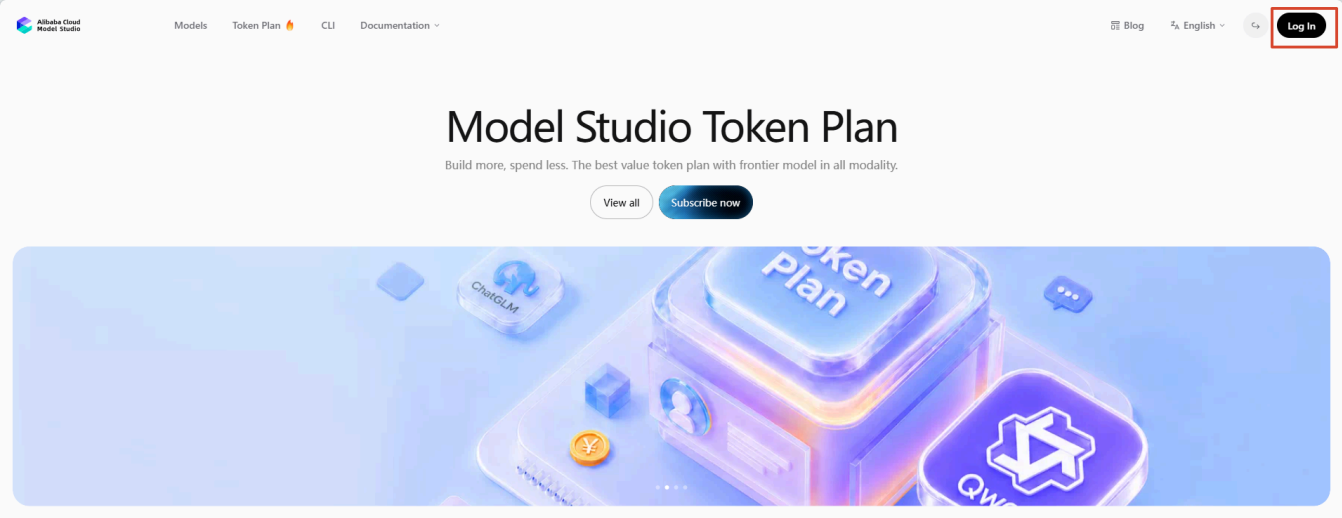
2. Hands-on operation

2.1 Obtaining the API credentials

1. **Log in to Alibaba Cloud**: visit Alibaba Cloud Model Studio Console and log in to your account; if the link is no longer valid, search for the "Alibaba Cloud Model Studio" platform yourself.
2. **Activate the Bailian service**: search for and enter the "Alibaba Cloud Model Studio" and activate the service following the instructions.
3. **Create an API KEY**: on the "API-KEY management" page of the Bailian platform, create a new API Key and **copy and save it immediately**, because it is only shown once.

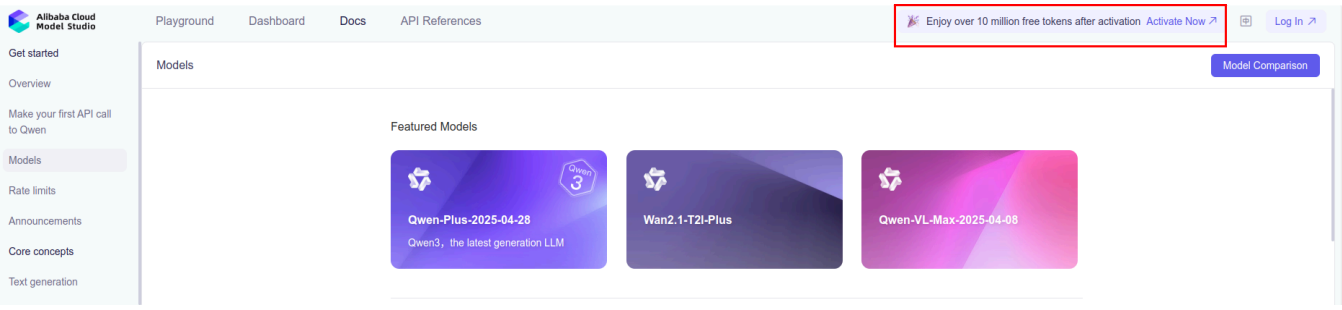
2.2 Step-by-Step Process

1. Register an account or log in



2. A newly registered user has to complete the real-name verification before it can claim the free quota.

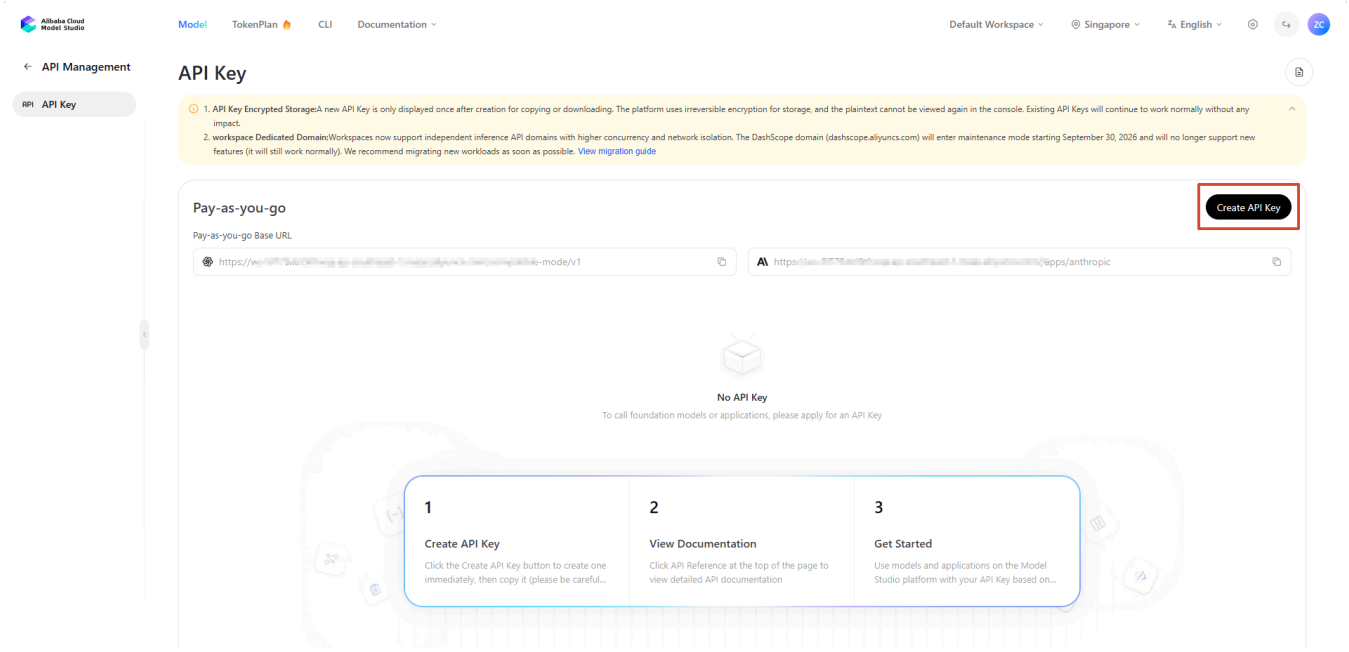
3. After completing real-name verification, return to the official website page and click to claim your free quota.



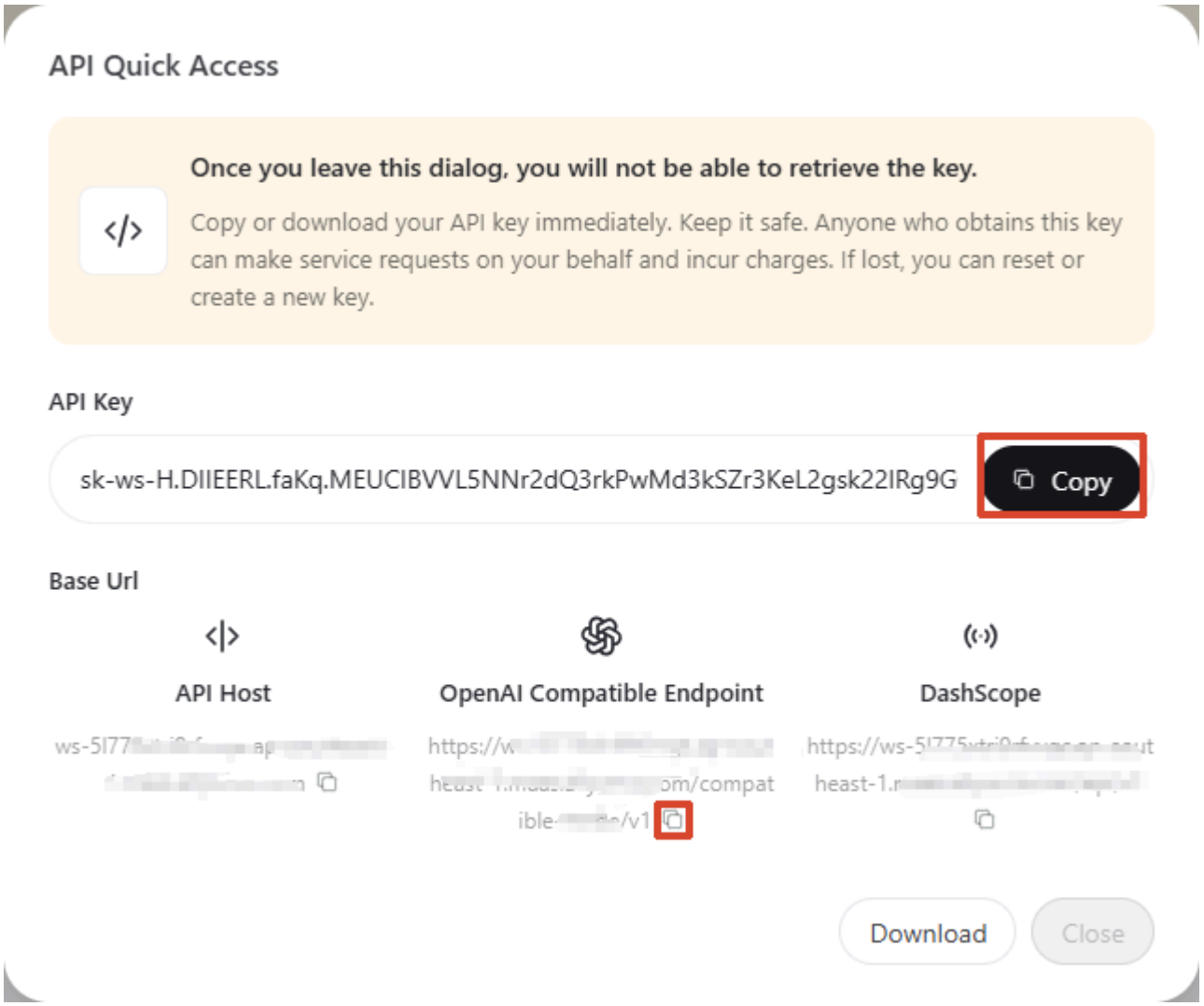
Tick that you have read and agreed, then confirm the activation and claim the free quota

4. Create the API KEY

Click the Alibaba Cloud Model Studio Console link, then click "Create API Key", select the default business space and click OK to finish the creation

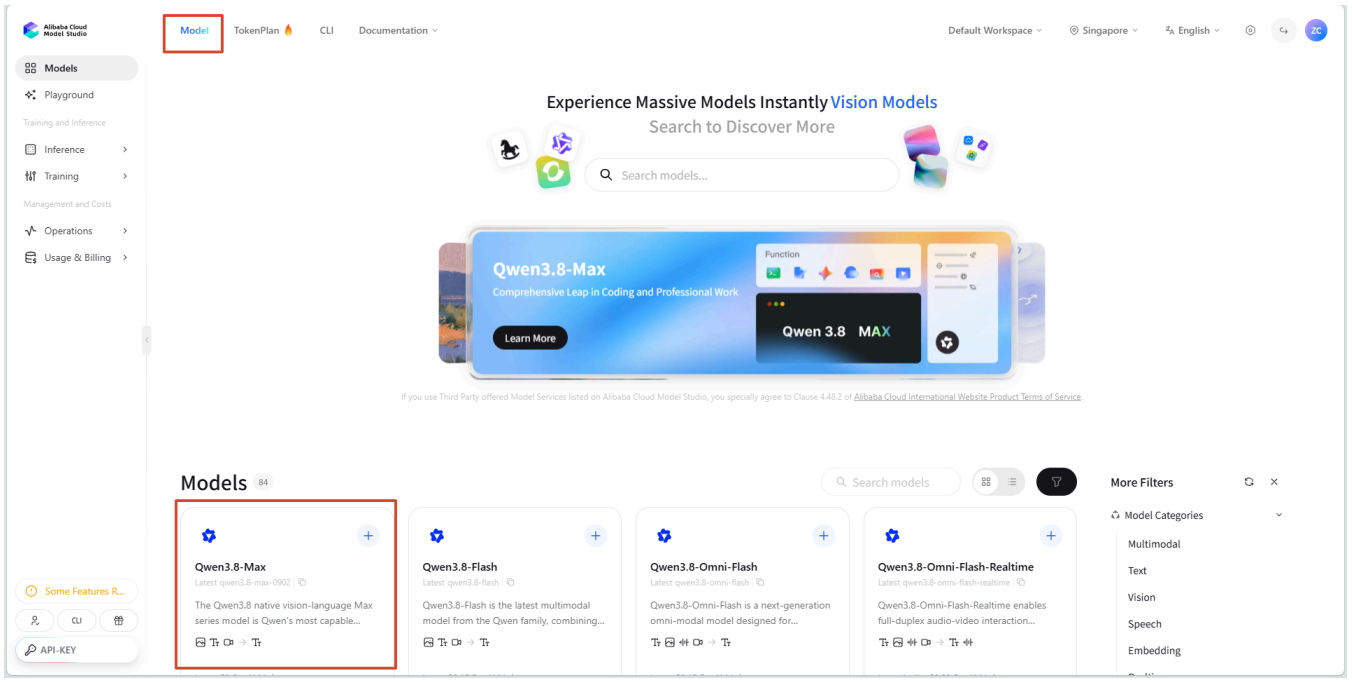


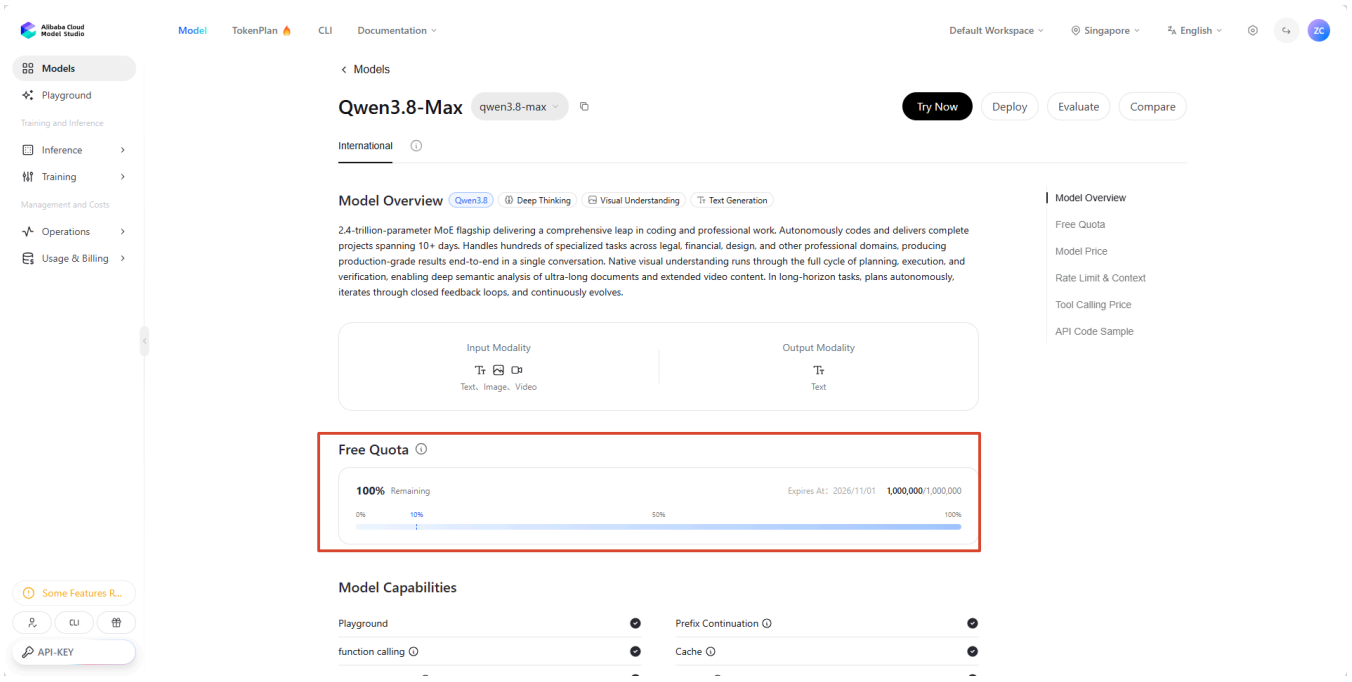
5. Copy your own API-KEY and OpenAI Compatible Endpoin



Then go to the configuration file and replace it with the KEY and Base URL you copied.

6. In the model plaza, click the three horizontal lines on the right to see the invocation name of each model and the remaining quota





2.3 Modifying the configuration file

1. **Open** `large_model_interface.yaml`:

```
vim ~/yahboom_ws/src/largemodel/config/large_model_interface.yaml
```

Then replace the sk-xxx in tongyi_api_key with the key you just copied.

tongyi_model configures the large model used for the conversation.

tongyi_media_model configures the large model used for text-to-image generation.

tongyi_base_url configures the compatible address of your own account. Fill in the OpenAI-compatible address you just copied.

```
# Tongyi platform configuration
tongyi_api_key: "sk-xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx"  # API key, paste your own key
tongyi_base_url: "https://dashscope.aliyuncs.com/compatible-mode/v1"  # Interface
address (the user has to configure the interface address of their own account)
tongyi_model: "qwen-vl-max-latest"  # The model in use, for example "qwen-vl-max",
"qwen-turbo"
tongyi_media_model: "flux-schnell"  # The Tongyi Qianwen text-to-image model
```

2. **Open** `yahboom.yaml`:

```
vim ~/yahboom_ws/src/largemodel/config/yahboom.yaml
```

3. **Switch the platform**: change the `llm_platform` parameter to `tongyi`.

```
model_service:
  ros__parameters:
    llm_platform: 'tongyi' # Change this from 'ollama' or another platform
```

Once the platform parameter has been switched, go back to the workspace and build again for the change to take effect.